Northwind Sales & Customer Analysis

SQL Data Analysis Case Study

Data Analytics Portfolio- Project 2

Tools: SQL Server • T-SQL • SQL Server Management Studio

Database: Northwind

A practical analysis of customers, sales, products, markets, employees, and shipping performance using relational business data.

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OVERVIEW

Tool Used: SQL Server (SSMS)

Dataset: Northwind Database

Records Analyzed: 91 Customers | 830 Orders | 2,155 Order Details

Categories: 8 | Employees: 9 | Countries: 21

SUMMARY

This project analyzes the Northwind database using SQL Server to uncover insights into customer behaviour, sales performance, product performance, regional markets, employee performance, and shipping efficiency. 10 SQL analyses were conducted across multiple relational tables to answer business questions and translate the results into business-focused findings.

Findings:

• The USA has the largest customer base and generates the highest country-level revenue.

• Quick-Stop in Germany is the highest-revenue customer, generating approximately $110,277.30.

• Beverages is the highest-revenue product category.

• Q4 (October–December) records the strongest trading activity in the analyzed order data.

• Margaret Peacock generates the highest employee revenue at approximately $232,890.85.

• 30 of 91 customers (33%) are above average in customer revenue based on the analysis.

BUSINESS PROBLEM

Northwind contains information about customers, orders, products, employees, and shipping operations. The purpose of this analysis is to understand which customers and markets contribute most to revenue, identify strong-performing products and employees, examine sales patterns over time, and evaluate shipping performance.

DATASET OVERVIEW

The Northwind database is a fictional business dataset representing a food and beverage company. The analysis uses relational tables connected through customer, order, product, employee, category, and shipping identifiers. It contains transactional data from a global retail company including customer orders, product details, employee records and shipping information across 21 countries from 1996–1998.

TABLE	PURPOSE
Customers	Customer information and location
Orders	Order dates, customers, employees and shippers
Order Details	Products, quantities, unit prices and discounts
Products	Product information and category relationships
Categories	Product category information
Employees	Employee and sales representative information
Shippers	Shipping company information

Dataset scale used: 91 customers, 830 orders, and 2,155 order-detail records.

TOOLS & SQL TECHNIQUES

Tools: SQL Server, SQL Server Management Studio (SSMS), T-SQL, Northwind relational database.

SQL TECHNIQUES USED

- SELECT, WHERE, ORDER BY
- GROUP BY with SUM, COUNT, AVG, ROUND
- JOINS
- TOP N filtering
- DATEDIFF and DATENAME date functions
- Revenue calculations
- CTE (Common Table Expression)
- Subquery within WHERE clause

QUESTIONS & ANALYSIS

1 — Which countries have the largest customer base?

SQL approach: Customers were grouped by country and counted.

Finding: The USA has the largest number of customers, indicating a strong customer presence in the US market.

2 — Which customers generate the most revenue?

SQL approach: Customers were joined to Orders and Order Details; revenue was calculated using price, quantity and discount.

Finding: Quick-Stop in Germany is the highest-revenue customer, generating approximately $110,277.30.

3 — Which product categories generate the most revenue?

SQL approach: Products were connected to Categories and Order Details and revenue was aggregated by category.

Finding: Beverages is the highest-revenue product category.

4 — Which countries generate the most revenue?

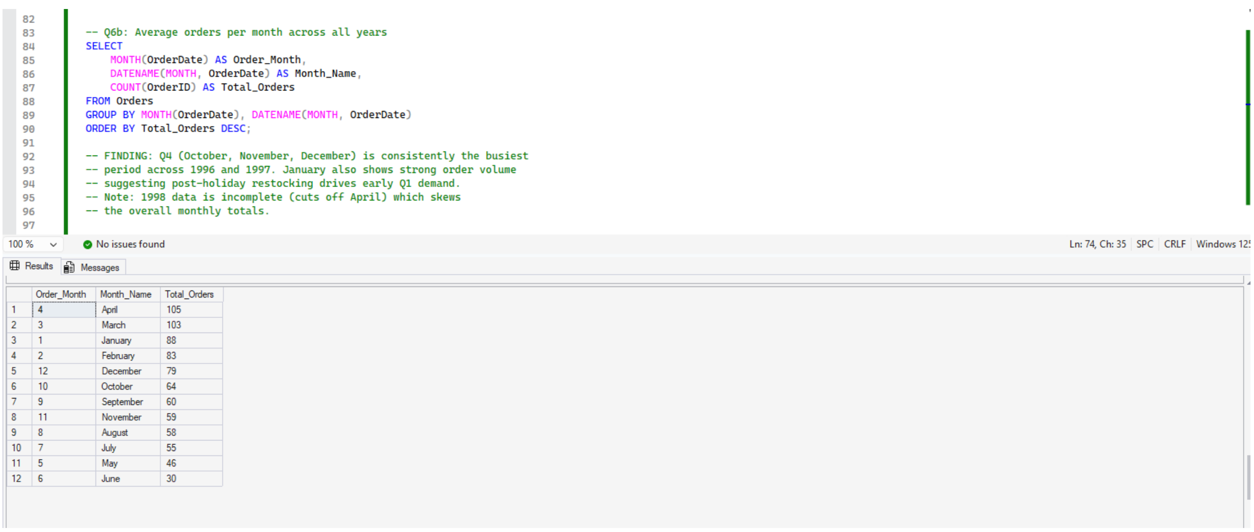
SQL approach: Customer, order and order-detail tables were joined and revenue was aggregated by country.

Finding: The USA generates the highest revenue among customer countries.

5 — How does order activity change over time?

SQL approach: Orders were grouped using date functions to examine monthly and quarterly activity.

Finding: Q4 (October–December) represents the strongest trading period in the analyzed order data.



```sql
-- Q6b: Average orders per month across all years
SELECT
    MONTH(OrderDate) AS Order_Month,
    DATENAME(MONTH, OrderDate) AS Month_Name,
    COUNT(OrderID) AS Total_Orders
FROM Orders
GROUP BY MONTH(OrderDate), DATENAME(MONTH, OrderDate)
ORDER BY Total_Orders DESC;

-- FINDING: Q4 (October, November, December) is consistently the busiest
-- period across 1996 and 1997. January also shows strong order volume
-- suggesting post-holiday restocking drives early Q1 demand.
-- Note: 1998 data is incomplete (cuts off April) which skews
-- the overall monthly totals.
```


	Order_Month	Month_Name	Total_Orders
1	4	April	105
2	3	March	103
3	1	January	88
4	2	February	83
5	12	December	79
6	10	October	64
7	9	September	60
8	11	November	59
9	8	August	58
10	7	July	55
11	5	May	46
12	6	June	30

6 — Which employees generate the most revenue?

SQL approach: Employees were joined to Orders and Order Details, with revenue aggregated by employee.

Finding: Margaret Peacock generates the highest employee revenue at approximately $232,890.85.

```sql
-- Q9: Employee performance by orders and revenue
SELECT
    E.FirstName + ' ' + E.LastName AS Employee_Name,
    COUNT(DISTINCT O.OrderID) AS Total_Orders,
    ROUND(SUM(OD.UnitPrice * OD.Quantity * (1 - OD.Discount)), 2) AS Total_Revenue
FROM Employees E
JOIN Orders O ON E.EmployeeID = O.EmployeeID
JOIN [Order Details] OD ON O.OrderID = OD.OrderID
GROUP BY E.FirstName, E.LastName
ORDER BY Total_Revenue DESC;
-- FINDING: Margaret Peacock is the top performing employee handling
-- 156 orders and generating $232,890.85 in revenue, leading both
-- in order volume and total revenue generated.
```

100 % No issues found Ln: 103, Ch: 83 SPC CRLF Windows 1

Results Messages

	Employee_Name	Total_Orders	Total_Revenue
1	Margaret Peacock	156	232890.85
2	Janet Leverling	127	202812.84
3	Nancy Davolio	123	192107.6
4	Andrew Fuller	96	166537.75
5	Laura Callahan	104	126862.28
6	Robert King	72	124568.24
7	Anne Dodsworth	43	77308.07
8	Michael Suyama	67	73913.13
9	Steven Buchanan	42	68792.28

7 — Which customers are above average in spending?

SQL approach: A customer-revenue CTE was compared with average customer revenue.

Finding: 30 of 91 customers (33%) are above average in customer revenue.

```sql
-- Q10: High value customers above average spend using a CTE
WITH CustomerRevenue AS (
    SELECT
        C.CompanyName,
        C.Country,
        ROUND(SUM(OD.UnitPrice * OD.Quantity * (1 - OD.Discount)), 2) AS Total_Revenue
    FROM Customers C
    JOIN Orders O ON C.CustomerID = O.CustomerID
    JOIN [Order Details] OD ON O.OrderID = OD.OrderID
    GROUP BY C.CompanyName, C.Country
)
SELECT
    CompanyName,
    Country,
    Total_Revenue
FROM CustomerRevenue
WHERE Total_Revenue > (SELECT AVG(Total_Revenue) FROM CustomerRevenue)
ORDER BY Total_Revenue DESC;
-- FINDING: 30 out of 91 customers (33%) are classified as high value,
-- spending above the average customer revenue. Quick-Stop remains the
-- top company.
-- These 30 customers should be the focus of retention strategies.
```

100 % No issues found Ln: 166, Ch: 16 SPC CRLF Windows 12

Results Messages

	CompanyName	Country	Total_Revenue
1	QUICK-Stop	Germany	110277.3
2	Ernst Handel	Austria	104874.98
3	Save-a-lot M...	USA	104361.95
4	Rattlesnake ...	USA	51097.8
5	Hungry Owl A...	Ireland	49979.91
6	Hanari Carnes	Brazil	32841.37
7	Königlich Ess...	Germany	30908.38
8	Folk och fä HB	Sweden	29567.56
9	Mère Paillarde	Canada	28872.19
10	White Clover ...	USA	27363.61
11	Frankenversa...	Germany	26656.56
12	Queen Cozinha	Brazil	25717.5
13	Berglunds sn...	Sweden	24927.58

8— Which shipping companies have the fastest average delivery times?

SQL approach: Orders were joined to Shippers and DATEDIFF was used to calculate shipping time.

Finding: Federal Shipping has the fastest average shipping time in the analyzed results, at approximately 1.7

days.

9 — What additional sales and customer patterns can be identified?

SQL approach: Customer, product, geographic, employee and order-level patterns were compared.

Finding: Revenue is concentrated among particular markets, customers, product categories and employees.

10 — What business opportunities emerge from the analysis?

SQL approach: The findings were considered together to identify commercially relevant actions.

Finding: Customer retention, market focus, inventory planning, seasonal preparation and logistics monitoring are key opportunities.

MY PROCESS

- Restored and explored Northwind database across 6 related tables
- Wrote 10 SQL queries of increasing complexity in SQL Server
- Used JOINs across 3 tables, GROUP BY aggregations, DATEDIFF
- date functions, TOP N filtering, and a CTE with subquery
- Identified revenue drivers across customers, products,
- categories, countries and employees

KEY FINDINGS

Customer Base	USA has the largest customer base.
Revenue by Market	USA generates the highest country-level revenue.
Top Customer	Quick-Stop - approximately $110,277.30.
Top Category	Beverages.
Sales Trend	Q4 records the strongest trading activity.
Top Employee	Margaret Peacock at approximately $232,890.85.
High- Value Customers	30 of 91 customers (33%) are above average.
Shipping	Federal Shipping has the shortest average shipping time in the analyzed results.

RECOMMENDATION

The business should prioritise retention of its 30 high-value customers who drive the majority of revenue, with Quick-Stop and Ernst Handel requiring dedicated account management. Shifting more orders to Federal Shipping would improve delivery times, reliability and customer satisfaction. The Grains/Cereals category should be reviewed for consolidation or replacement with higher performing product lines. Finally, Q4 stock and staffing levels should be increased to capitalise on the consistently high seasonal demand.

LIMITATIONS

This analysis is based on the Northwind sample database and represents a fictional business environment. The dataset covers a limited period and may not reflect current market conditions. Factors such as operating costs, marketing spend, customer acquisition cost and profitability were not included.

Revenue should not be interpreted as profit; additional financial data would be required to assess profitability.

CONCLUSION

The Northwind analysis demonstrates how SQL can transform relational business data into meaningful insights. The project identified important customer segments, high-performing markets and product categories, sales trends, employee performance, and shipping patterns while combining technical SQL skills with business-oriented interpretation.

The complete SQL queries are provided separately in **customer_analysis.sql**. The SQL file contains the full query logic and findings for the analyses in this case study.

Project files: README.md • customer_analysis.sql • Customer_Analysis_CaseStudy.pdf